FIREWATCH AI

Wildfire Risk Prediction & Impact Assessment

Northern Algeria · 2018–2024 · 3-day fire-occurrence horizon

Academic data-mining project
Prediction-first interface · reproducible evaluation · geospatial impact assessment

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EXECUTIVE SUMMARY

What this report covers

This report documents the prediction-first redesign of the FIREWATCH AI dashboard, the reproducible evaluation of the deployed model, and the supporting artifacts that were added to the project. All numbers in the tables below are produced by **notebook.py** and stored in **report/evaluation_summary.json**; the figures are the exact PNG files written to **report/figures/**.

0.2441 PR-AUC	**0.8812** ROC-AUC	**0.03618** BRIER SCORE	**50.5%** FIRES IN TOP DECILE
0.125 BRIER SKILL SCORE	**0.0051** CALIBRATION ERROR	**0.88** CALIBRATION SLOPE	**4.32%** TEST BASE RATE

Work delivered

- **Prediction-first landing page** (**app.py**): the app now opens on a guided *enter conditions → get a risk estimate* flow, with scenario presets, a probability gauge, an explanation chart and actionable guidance.
- **Testable prediction core** (**dashboard/predictor.py**): Streamlit-free feature building, scoring, sensitivity analysis and guidance helpers, covered by **tests/test_predictor.py**.
- **Evaluation notebook** (**notebook.py**): reproduces the chronological evaluation, prints statistics and metrics, and writes ten figures plus machine-readable results.
- **Documentation**: a complete project README and this PDF report.
- **Regression safety**: the full test suite (249 tests) passes after the changes, and the SHA256-pinned model card was left untouched.

Headline result

On the chronological hold-out (2022–2024, 228,437 cell-days, 4.32% base rate) the calibrated model reaches a PR-AUC of **0.2441** against **0.2066** for a per-cell, per-month climatology baseline. It captures **50.5%** of all observed fires inside the highest-risk 10% of cell-days, while remaining well calibrated (Brier 0.03618, ECE 0.0051, skill score 0.1253).

Report map

Section	Content
01 · Product	The prediction-first interface: journey, presets, supporting tools and design decisions
02 · Model & data	Artifact, feature contract, dataset and chronological split
03 · Evaluation	Test-period metrics, risk bands and year-by-year performance
04 · Evidence	The ten figures produced by notebook.py
05 · Interpretation	What drives the score, and the limits of that reading
06 · Reproducibility	Commands to rebuild every number and artifact

01 PRODUCT

The prediction-first interface

The previous entry point was an analytics landing page: a visitor had to browse charts before discovering that the project can actually predict anything. The new entry point inverts that journey. The first thing a user sees is a short form and a result.

User journey

Step	What the user does	What the app returns
1 · Locate	Picks a latitude / longitude and a date (or uses the defaults).	The snapped grid cell actually used by the model is displayed.
2 · Describe	Chooses a scenario preset or sets six fire-weather values.	Values are echoed back as a readable condition summary.
3 · Predict	Presses **Predict fire risk**.	Calibrated probability, risk band, gauge and plain-language reading.
4 · Understand	Reads the sensitivity chart and the guidance list.	Which inputs push the score up or down, and what to do about it.

Design decisions

- **Prediction before analytics.** Charts remain available in the sidebar, but they no longer gate the first interaction.
- **Few inputs, honest defaults.** Six weather values and a location are enough; the 7/14/30-day lag features are derived from them and can be overridden in an advanced panel.
- **No silent imputation.** Every value sent to the model is shown to the user, and optional live NASA POWER data is clearly labelled as a fetch.
- **Explanation is local and transparent.** The sensitivity chart moves one feature at a time to a fire-suppressing reference; it is a perturbation study, not a causal claim.
- **Calibrated uncertainty.** The score is a probability, displayed with its climatological base rate so a user can judge whether it is unusual.

Scenario presets

Preset	Conditions	Typical use
Summer heatwave	43 °C peaks, 15% RH, 42 dry days	Peak-season stress test
Dry spring	27 °C, 46% RH, light rain, 4 dry days	Early-season watch
Mild autumn	22 °C, 58% RH, moderate soil moisture	Should stay low
Winter wet	14 °C, 84% RH, 16 mm rain	Saturated-fuel reference

Presets only prefill the form; every value remains editable. A manual configuration that scores unexpectedly is therefore always inspectable.

> **Scope of a prediction**
> A score estimates the probability that a satellite fire detection occurs in the selected grid cell within the next 3 days. The grid is coarse (~0.5°), so the output is a **regional warning signal**, not a forecast of a specific fire, and not an official emergency notification.

Beyond a single score

- **Scenario comparison.** A collapsed table and bar chart score all four presets on the same grid cell and date, so the model's response to weather can be read next to the historical base rate.

- **Stable results.** The result panel renders from a snapshot taken when the user presses Predict. Editing an input afterwards raises a warning that names the changed drivers instead of silently moving the displayed score.
- **Shareable links.** Location, date and scenario are written to the URL as **?lat=&lon=&date=&scenario=&run=1**. A link carrying **run=1** re-scores on arrival, which makes a result easy to discuss or cite.
- **JSON export.** A downloadable record captures the model version, grid cell, inputs, calibrated score, band and drivers for archiving or reporting.
- **Accessibility.** The shared design system provides visible keyboard focus rings on light and dark surfaces, announced callouts, readable disabled controls and a reduced-motion fallback.

02 MODEL & DATA

Model, features and dataset

The deployed artifact is model **v0.1.0**, a calibrated gradient-boosting classifier over **20 features**. The target is the occurrence of an observed fire in the cell within the next 3 days.

Dataset

Property	Value
Study area	Northern Algeria (32.0–37.5°N, -2.0–9.0°E)
Rows (cell-days)	521,246
Grid cells	209
Period	2018-03-02 → 2024-12-28
Positive cell-days	26,135 (5.01%)

Chronological split

Split	Definition	Rows
Train	Dates before 2021	**215,897**
Validation	Year 2021 (calibration)	75,658
Test	2022 – 2024 inclusive	228,437

Feature set

Feature	Feature
lat_cell	rhmin_7d
lon_cell	rain_14d
T2M	tmax_14d
T2M_MAX	rhmin_14d
RH2M	rain_30d
WS2M	tmax_30d
PRECTOTCORR	rhmin_30d
GWETROOT	dry_days
rain_7d	doy_sin
tmax_7d	doy_cos

Feature order is authoritative in models/model_card.json and is enforced by the test suite.

Model-card drift

The shipped card records PR-AUC 0.1582 and a 2.37% base rate, while this evaluation obtains 0.2441 at a 4.32% base rate. The card was written on an earlier panel snapshot. It is intentionally left unchanged because an integrity test pins its SHA256; this report is the up-to-date record.

03 EVALUATION

Test-period results

Fires are rare, so accuracy is not informative. The model is judged on ranking quality (PR-AUC, ROC-AUC, top-decile capture), on calibration (Brier score, skill score, ECE) and against a climatology baseline that already knows where and when fires usually happen.

Metric	Value	Reading
PR-AUC (calibrated)	**0.2441**	vs 0.2066 climatology
PR-AUC (raw scores)	0.2529	before calibration
ROC-AUC	0.8812	strong separation
Brier score	0.03618	lower is better
Brier skill score	0.1253	0 = base-rate forecast
Expected calibration error	0.0051	10 quantile bins
Calibration slope	0.881	1.0 = perfect
Fires in top decile	50.5%	of all observed fires
Base rate (test)	4.32%	random PR-AUC
Precision @ 0.5	51.4%	recall 4.2%
Best F1	0.308	at threshold 0.155

Risk bands

Band	Cell-days	Mean score	Observed rate	Share of fires
LOW	**107,903**	**0.08%**	**0.12%**	**1.3%**
MODERATE	33,587	1.09%	1.22%	4.2%
HIGH	36,300	4.15%	4.07%	14.9%
VERY HIGH	50,647	17.48%	15.53%	79.6%

Each band behaves monotonically: the observed fire rate rises from 0.12% in LOW to 15.53% in VERY HIGH, and the top band contains most of the fires.

Performance by year

Year	Rows	Base rate	PR-AUC model	PR-AUC climatology	Brier	Fires in top decile
2022	**76,285**	**4.77%**	**0.2960**	**0.2540**	**0.0379**	**54.5%**
2023	76,285	4.72%	0.2607	0.2175	0.0388	50.1%
2024	75,867	3.47%	0.1732	0.1510	0.0318	44.4%

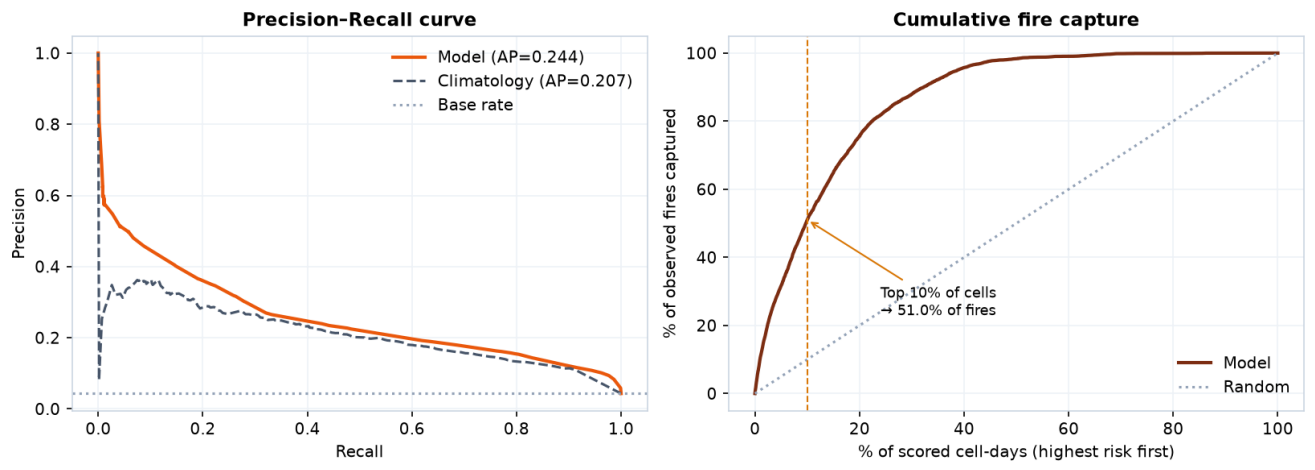
2024 is visibly harder: the base rate falls to 3.47%, and so does the achievable PR-AUC. The model still beats the climatology baseline in every test year.

04 EVIDENCE

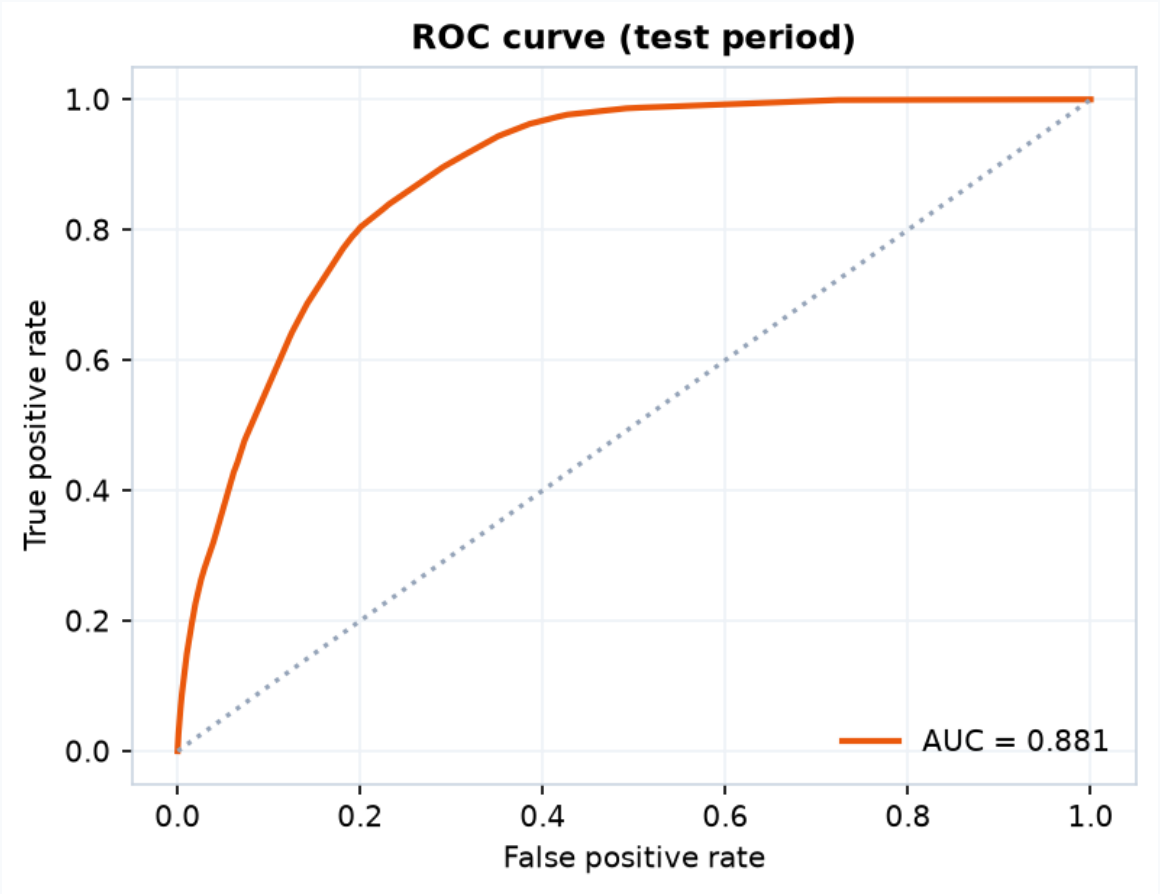
Figures

All figures are generated by **python notebook.py**. They use the test period only, except the seasonality and data-overview panels which describe the full panel.

Ranking quality

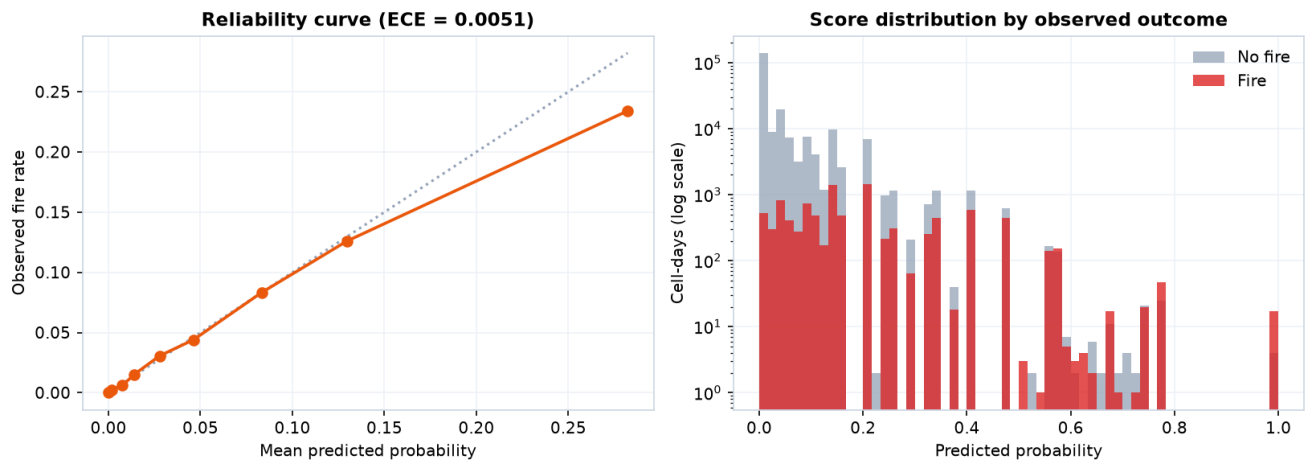


Precision-recall curve against the base rate, with the fire-capture curve. The model concentrates observed fires in a small share of alerted cell-days.

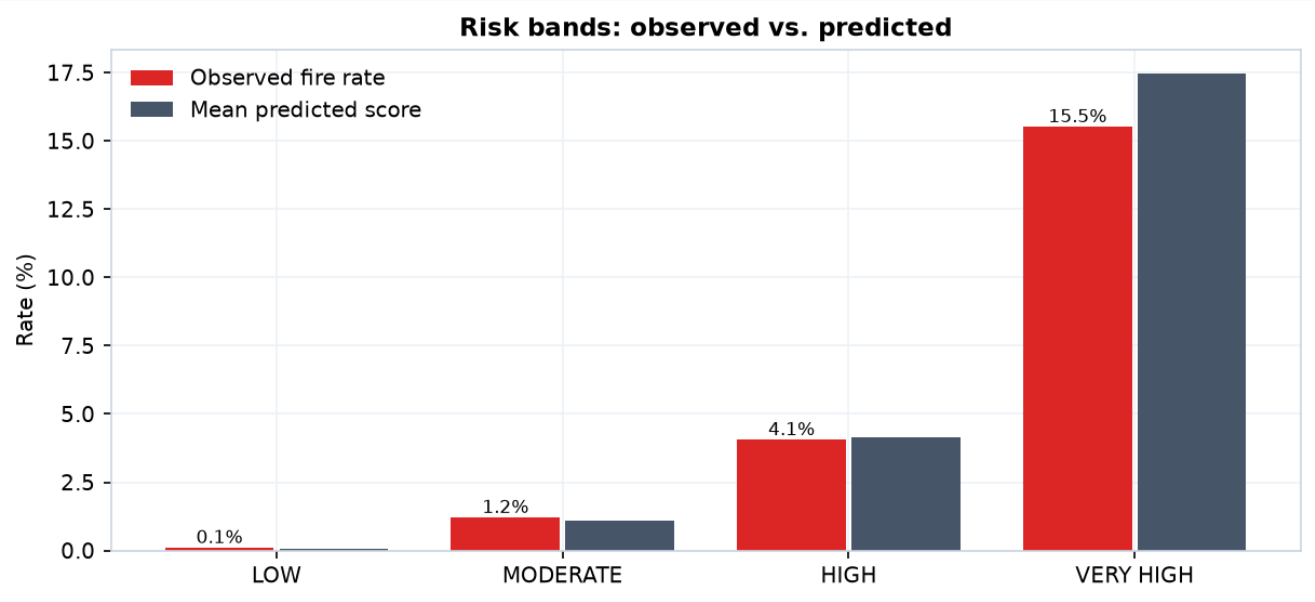


ROC curve on the 2022–2024 hold-out.

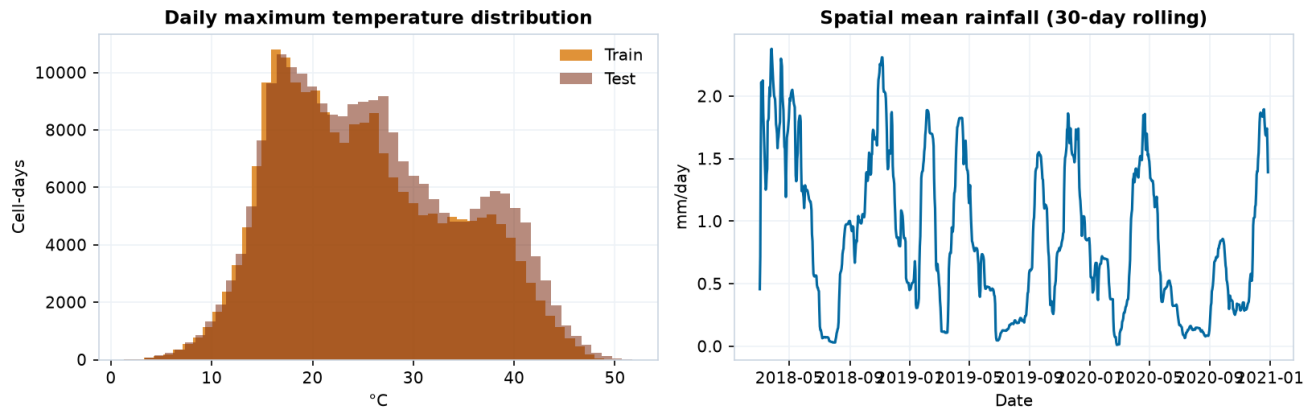
Calibration, bands and data



Reliability diagram. Predicted probabilities track observed frequencies closely (ECE ≈ 0.005), which is what makes the gauge meaningful in the UI.

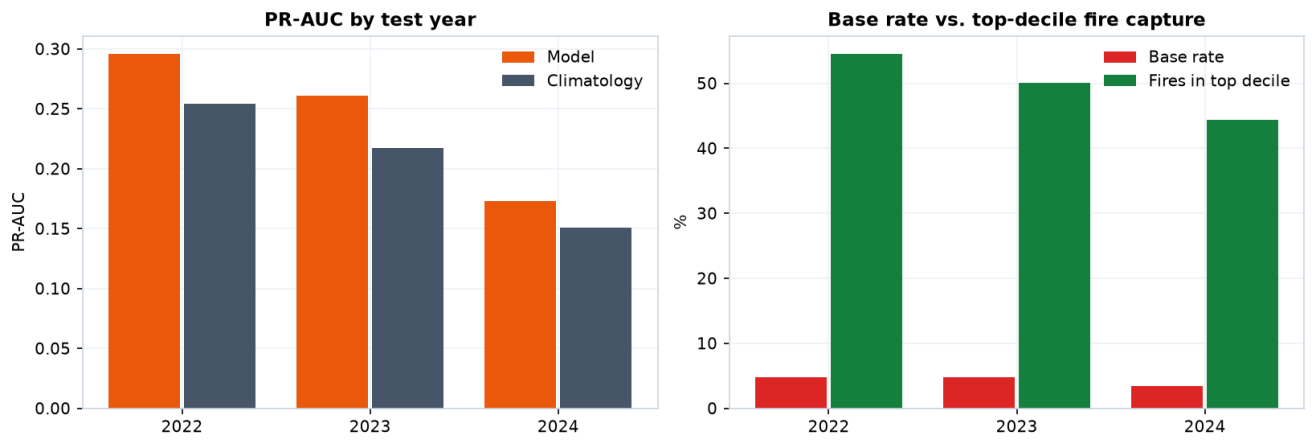


Risk-band calibration: mean predicted score versus observed fire rate per band.

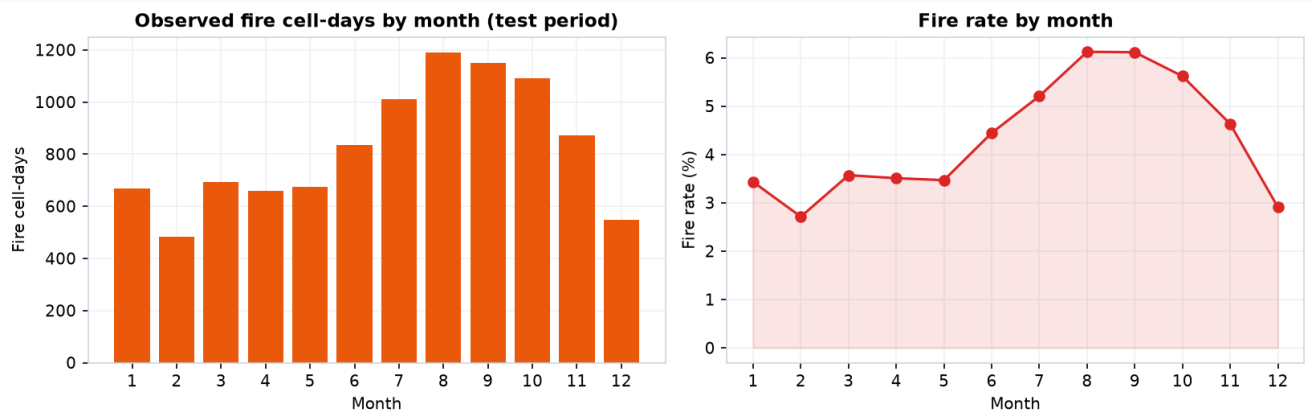


Dataset overview: rows per year, positive rate and cell coverage.

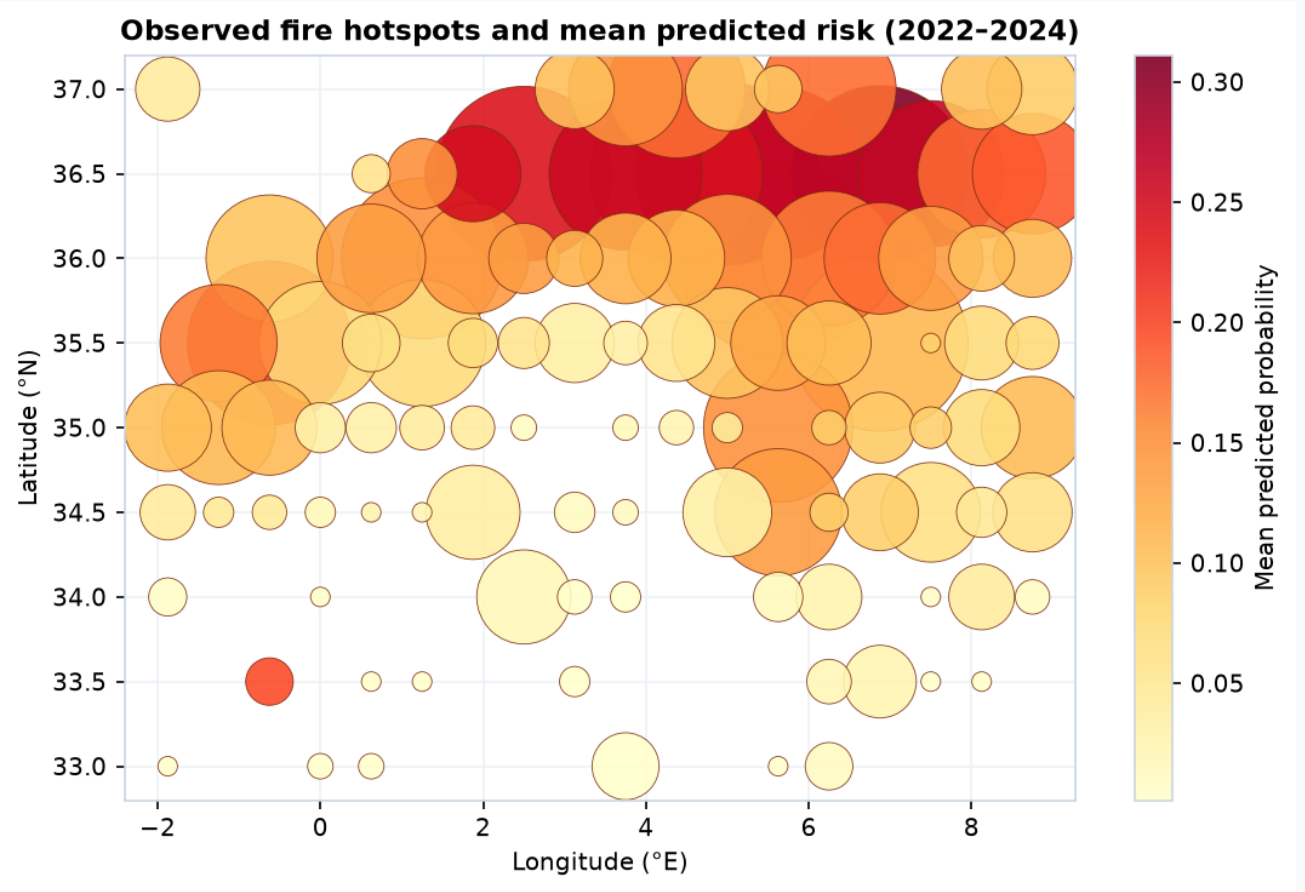
Temporal and spatial behaviour



PR-AUC and top-decile capture per test year, against climatology.

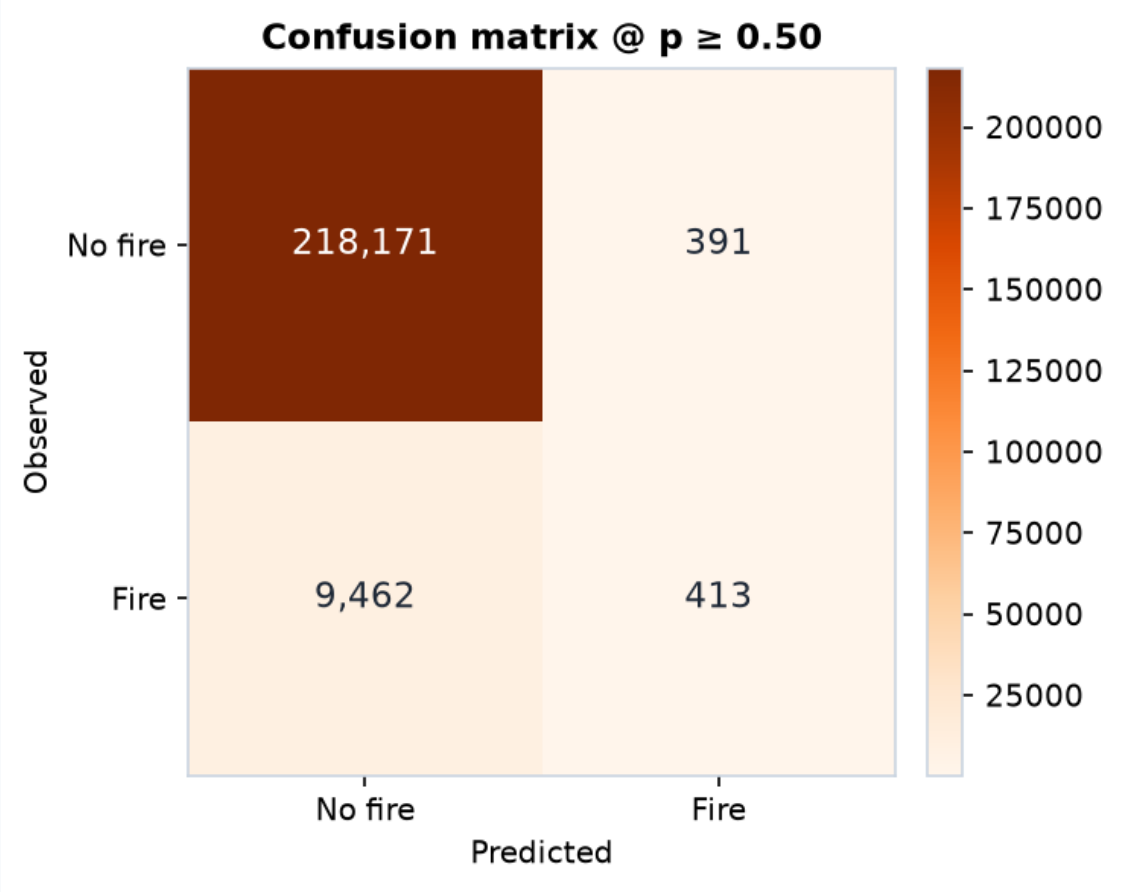


Fire seasonality across the full panel: the summer peak the model must learn.

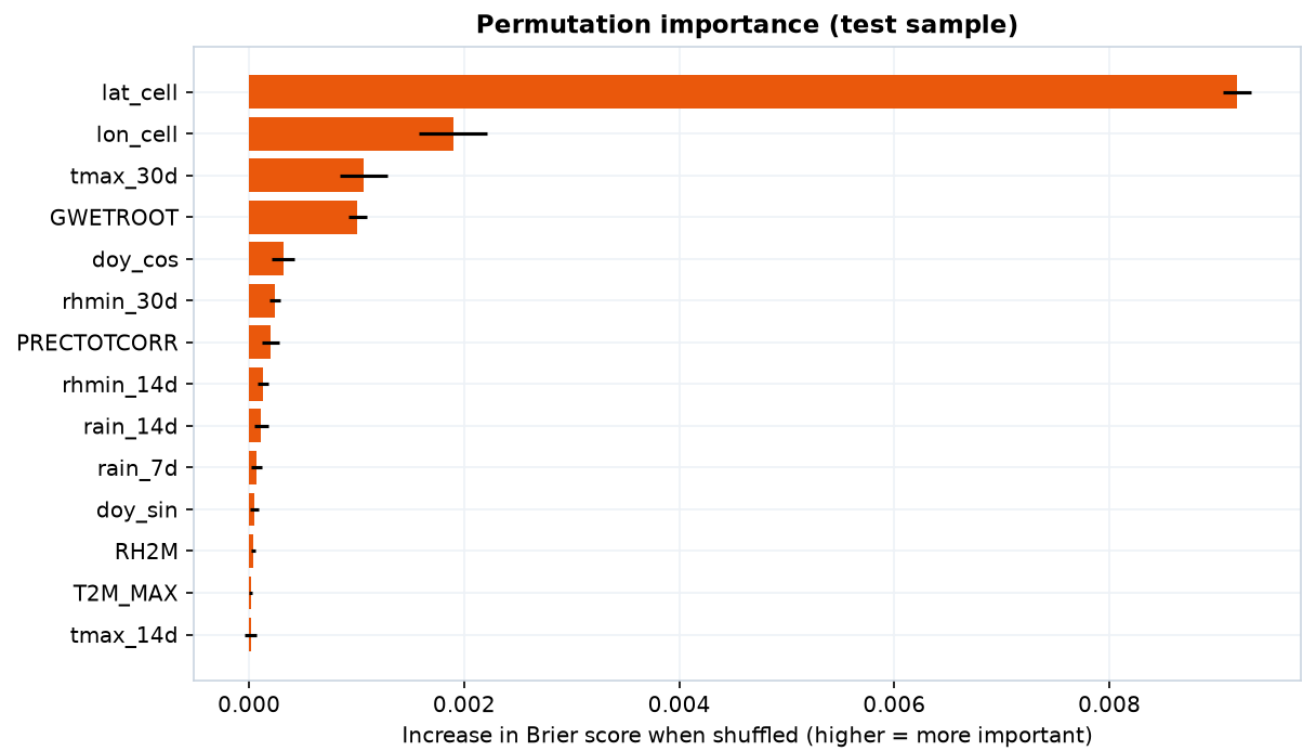


Spatial hotspot map of observed fires, showing why grid-cell coordinates are strong predictors.

Errors and feature attribution



Confusion matrix at p ≥ 0.5. With a 4.3% base rate this threshold is deliberately conservative; the operational value lies in ranking, not in hard classification.



Permutation importance for all 20 features. Coordinates dominate, followed by 30-day heat and root-zone soil wetness.

05 INTERPRETATION

What drives the score

Sampled permutation importance is dominated by the cell coordinates. That is an honest and important finding: the model learns *where* fires happen very well, and adds weather information on top of that geography. Users should read the score as a location-aware seasonal risk, not as a pure weather index.

Feature	Importance (mean)	Std
lat_cell	**0.00918**	**0.00013**
lon_cell	0.00190	0.00032
tmax_30d	0.00106	0.00022
GWETROOT	0.00101	0.00008
doy_cos	0.00032	0.00011
rhmin_30d	0.00024	0.00005
PRECTOTCORR	0.00020	0.00008
rhmin_14d	0.00013	0.00005
rain_14d	0.00011	0.00007
rain_7d	0.00007	0.00005
doy_sin	0.00005	0.00004
RH2M	0.00004	0.00002

Top 12 of 20 features. Full ranking in report/evaluation_summary.json; the bar chart of all 20 appears in section 04.

Limitations to state explicitly

- **Coarse grid.** NASA POWER cells are ~0.5°, so local terrain and microclimate are invisible to the model.
- **Location dominance.** Most predictive signal is geographic; a single day's weather moves the score less than the cell does.
- **Missed detections.** FIRMS can miss fires under cloud cover or between satellite overpasses, which adds label noise.
- **Regional scope.** Trained on northern Algeria; transfer to other regions is untested.
- **Not an emergency system.** Scores support analysis and prioritisation; they are not official forecasts.

06 REPRODUCIBILITY

How to reproduce everything

Every result in this report is regenerated from the repository with four commands. No manual editing of numbers is involved.

Setup

```
python -m venv .venv
.venv\Scripts\activate (Windows)
python -m pip install -r requirements.txt
```

Evaluate, then build the report

```
python notebook.py
python report/build_report.py
```

Launch the application

```
streamlit run app.py
```

Verify the whole project

```
python -m pytest -q
```

Artifacts produced

Artifact	Purpose
report/EVALUATION.md	Human-readable evaluation report
report/evaluation_summary.json	Machine-readable metrics and tables
report/figures/*.png	Ten evaluation figures used in this PDF
report/FIREWATCH_AI_Report.pdf	This document
report/build_report.py	Deterministic PDF generator

Integrity note
models/model_card.json is protected by a SHA256 assertion in tests/test_impact_integrity.py and was not modified. The metrics in this report come from re-scoring the deployed artifact, so they describe the current code and data rather than the historical card values.

FIREWATCH AI — Data-driven wildfire risk analysis and geospatial impact assessment.